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Sales Data Analysis for Business Solutions

Advanced Topics In Analytics

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1. Executive Summary

The modern electronics retail landscape is defined by an unprecedented convergence of complexity: shifting consumer expectations, razor-thin product margins, and an ever-expanding volume of transactional data generated at every touchpoint of the customer journey. In this environment, operational intuition and static business rules are no longer sufficient to sustain competitive advantage. The organizations that thrive are those capable of transforming raw data into precise, actionable intelligence, leveraging the full spectrum of analytical methods to guide strategic decisions across logistics, pricing, and customer relationship management.

This report presents a comprehensive analytical investigation of 20,000 transactional records sourced from an electronics retailer, spanning a twelve-month period. By applying a suite of advanced analytical techniques, including multi-objective genetic optimization, log-log regression modeling, unsupervised machine learning, and statistical hypothesis testing, this report seeks to uncover the structural inefficiencies embedded within the company's current operations and translate them into concrete, prioritized recommendations.

Three core research questions guide the analysis. The first examines how the company's shipping fulfillment strategy can be optimized to simultaneously minimize logistics costs and maximize customer satisfaction, two objectives that are inherently in tension. The second investigates the Price Elasticity of Demand (PED) across five major product categories, determining whether discounting strategies are capable of driving meaningful volume gains or whether they simply erode profitability without a corresponding behavioral response. The third research question evaluates the effectiveness of the existing loyalty program by segmenting the customer base into behaviorally distinct cohorts using RFM (Recency, Frequency, Monetary) analysis, and statistically testing whether loyalty membership translates into higher add-on spending.

Together, these questions address the three pillars of retail profitability: operational efficiency, pricing power, and customer lifetime value. The findings challenge several widely held assumptions in electronics retail management and lay the groundwork for a data-driven strategic realignment, one that replaces mass marketing and rigid operational policies with dynamic, micro-targeted approaches grounded in empirical evidence.

2. Research Questions

This report undertakes an exhaustive analysis of a proprietary electronics retail dataset containing transactional records spanning a one-year period from September 2023 to September 2024. The dataset provides a granular, multidimensional view of consumer purchasing behavior, capturing critical elements such as customer demographics, product typologies, pricing structures, loyalty membership status, add-on acquisition patterns, shipping preferences, and ultimate order fulfillment outcomes. By leveraging this data architecture, the analysis explores three fundamental research questions designed to optimize enterprise performance.

The first research question investigates how to optimize the company's shipping strategy to balance operational costs with customer satisfaction. Since customer service lead-time is a key differentiator in the electronic sales market, this research question helps construct a data-driven strategy for maximizing long-term profitability and operational efficiency in electronics retail.

The second research question explores the Price Elasticity of Demand (PED) across different SKUs in the same product type. This helps reveal which SKUs will drive volume when discounted, and which will merely erode margins with no volume gain. From that, the marketing team can decide whether it would offer discounts for a specific high-end SKU.

The third research question focuses on segmenting the customer base according to empirical purchasing behavior and evaluating loyal members' spending on add-on. This helps the company find out whether cancellation rate differs across these newly discovered clusters and allocate marketing resources effectively.

By answering these questions, this analysis aims to provide precise, data-driven recommendations that enable the company to optimize its operational and marketing efforts. The inferences gained help identifying key drivers for company profitability and allow more strategic targeting by tailoring product offers to specific customer segments.

3. Methodology & Problem Formulation

The foundation of any robust optimization framework is a deep, structural understanding of the underlying data architecture. The utilized dataset comprises 20,000 of individual transactional records between September 2023 and September 2024, each characterized by 16

distinct attributes that map the end-to-end customer journey from initial selection to final order disposition.

All analyses were implemented using Google Colab, a free, cloud-based Jupyter Notebook environment provided by Google, allowing users to write and execute Python code through the browser. Raw transactional data is not suitable for immediate ingestion by machine learning algorithms or optimization solvers. Therefore, a systematic data transformation process is executed within the Python environment. The Add-ons Purchased field contains variable-length, comma-separated text strings representing multiple simultaneous purchases. In Python, this feature must be parsed and transformed. This process converts the text string into a sparse matrix of binary features, creating distinct, quantifiable columns for Has_Accessory, Has_Extended_Warranty, and Has_Impulse_Item. Furthermore, missing values within this column, representing transactions with zero ancillary purchases, must be carefully imputed to avoid algorithmic errors, ensuring the corresponding Add-on Total is strictly cast as a 0.0 float value.

Temporal features also require extensive manipulation. The Purchase Date variable must be decomposed using pandas datetime functionality to extract cyclical features such as the month of the year, and binary flags indicating each specific month. This temporal extraction is vital for capturing the seasonality that is endemic to the electronics retail sector.

Categorical variables, such as Product Type and Shipping Type, must be converted into numerical formats using one-hot encoding, expanding the dimensionality of the dataset while preserving the nominal relationships between categories. The target variable for operational modeling, Order Status, is transformed via label encoding, where a "Cancelled" order is mapped to the integer 1 and a "Completed" order is mapped to the integer 0, facilitating binary classification tasks.

3.1. Shipping Fulfillment Optimization (RQ1)

The goal of this question is to optimize its shipping strategy to balance operational costs with customer satisfaction. To address this, the research objectives must first be clarified:

- Objective 1: **Minimize shipping costs:** (Assuming weights: Standard is lowest cost, Same Day is highest).

$$\text{Minimize } Z_1 = \sum_{i=1}^n \sum_{j=1}^m C_j x_{ij}$$

(x_{ij} is a binary variable where $x_{ij}=1$ if order i is shipped using shipping method j , and

0 otherwise. C_j is the estimated cost of shipping method j).

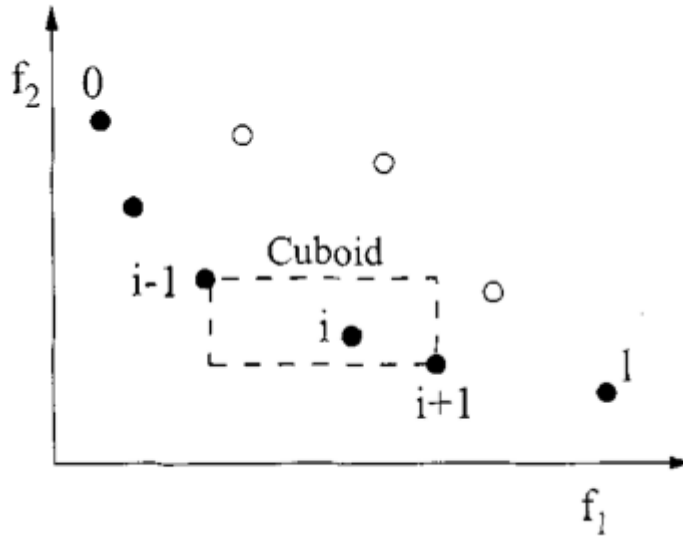
- Objective 2: **Maximize customer satisfaction:**

$$\text{Maximize } Z_2 = \sum_{i=1}^n \sum_{j=1}^m V_{ij} x_{ij}$$

(V_{ij} is the value score of giving order i shipping method j).

The problem was formulated as a multi-objective optimization task, recognizing the fundamental trade-off between logistics efficiency and service excellence as defined by Christopher (2016). To simulate a daily batch, 200 completed orders in the dataset were taken as a sample for the model.

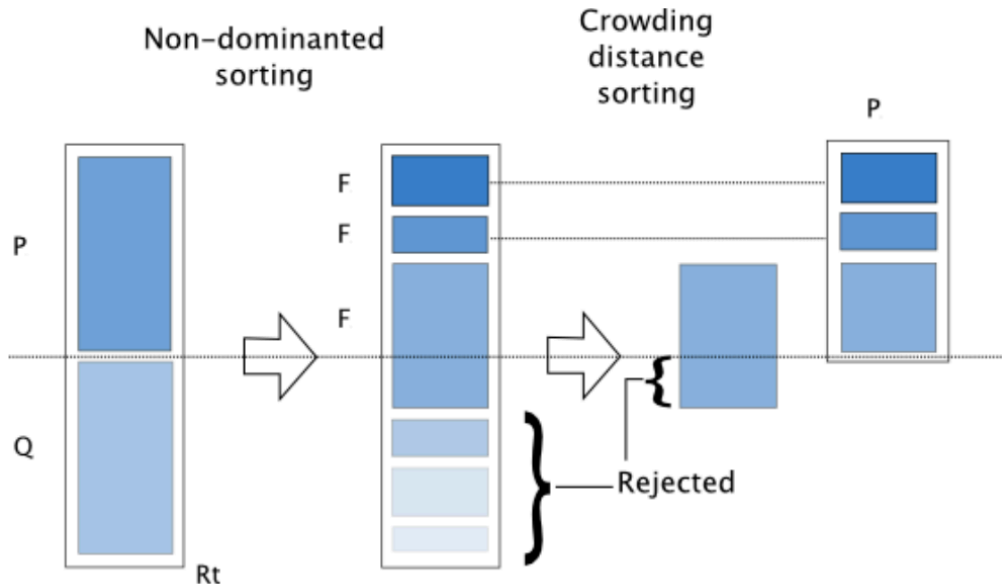
To navigate the Pareto Front (Deb et al., 2002), a genetic algorithm was implemented using the pymoo library in Python to find a non-dominated set of fulfillment solutions (NSGA-II). In this algorithm, individuals are selected frontwise. By doing so, there will be a situation where a front needs to be split because not all individuals are allowed to survive. In this splitting front, solutions are selected based on crowding distance.



Source: Deb et al., 2002

Fig. 1. Crowding distance calculation

The crowding distance is the Manhattan Distance in the objective space. However, the extreme points are desired to be kept every generation and, therefore, get assigned a crowding distance of infinity.



Source: pymoo.org

Fig. 2. NSGA-II procedure.

A combined population R_t of P and Q is formed. Then, the population R_t is sorted according to nondomination. Since all previous and current population members are included in R_t , elitism is ensured. Now, solutions belonging to the best non-dominated set F are the best solutions in the combined population and must be emphasized more than any other solution in the combined population. From this set of solutions, one specific optimal solution that is most balanced can be extracted using the `argmin()` function in the Numpy library for the crowding distance formula.

Heskett, J. L., et al. (1994) established the positive link between service quality, customer loyalty, and long-term profit. Therefore, a loyalty multiplier of 1.5 was set to favour loyalty members for premium shipping methods.

$$\text{satisfaction scores} = \text{base satisfaction} * \text{loyalty multipliers}$$

To make the model closer to a real-world problem, costs, warehouse capacity and base satisfaction scores were assumed for each of the shipping methods.

	Standard	Expedited	Express	Overnight	Same Day
Cost (\$)	5	15	25	30	50

Capacity	Unlimited	100	60	30	20
Satisfaction score	3	6	8	9	10

Table 1. Assumed constants

Before running the model, some constraints were set to ensure the feasibility of the result:

- The number of shipments cannot exceed daily warehouse capacity.
- All "Completed" status orders must be assigned a shipping method.

3.2. Price Elasticity of Demand across different SKUs (RQ2)

A specialized and critical sub-domain of sensitivity analysis in retail is the measurement of price elasticity. Price Elasticity of Demand (PED) is an economic metric that measures the responsiveness of consumer quantity demanded to fluctuations in a product's price (Mucollari et al., 2022).

Mathematically, the relationship between price (P) and quantity demanded (Q) can be modelled using a log-log linear regression model on historical sales data. Taking the natural logarithm of both sides of an exponential demand curve yields the linear equation:

$$\log(Q) = \beta_0 + e * \log(P) + \varepsilon$$

Here, the coefficient e directly represents the price elasticity of demand. To prevent severe bias, some control variables were added to the model. When looking at cross-sectional data, higher-priced items generally have better features or brand trust. If quality adjustment in cross-sectional demand analysis is ignored, the estimated price elasticity will be heavily biased (Chung, 2006). Therefore, "Rating" column was used to control the quality of the product. Also, the seasonal controls were retained to ensure demand spikes around holidays do not skew the baseline relationship between price and quantity. Furthermore, Loyalty Member status is a powerful moderating variable, making enrolled consumers substantially more price-inelastic due to switching costs and brand affinity. Ultimately, a standard Ordinal Least Square (OLS) regression analysis was implemented using Python's statsmodels library.

3.3. Customer Segmentation and Loyal Status Evaluation (RQ3)

In modern retail, mass marketing has been entirely superseded by data-driven, highly targeted micro-segmentation strategies. To effectively allocate marketing resources, the customer base

must be stratified according to empirical purchasing behavior. This is achieved through Recency, Frequency, and Monetary (RFM) analysis, a foundational marketing analytics framework (Tang, 2025). Utilizing the dataset, Recency is computed as the temporal distance between the anchor date of the dataset and the customer's maximum Purchase Date. Frequency is derived from the aggregate count of unique Order ID instances per Customer ID, and Monetary value is established by summing the Total Price and Add-on Total. By standardizing these RFM metrics and applying unsupervised machine learning algorithms K-Means clustering implemented via Python's scikit-learn, the customer base is partitioned into actionable strategic segments. The algorithm iteratively assigns customers to clusters by minimizing the within-cluster sum of squares (WCSS), identifying distinct segments. Parallel to segmentation is the critical evaluation of the Loyalty Member variable. Loyalty programs are no longer viewed as peripheral promotional tactics; they are central strategic levers designed to maximize Customer Lifetime Value (CLV) and insulate the brand from intense price-based competition (Kaur, 2024). A Welch's T-Test was implemented via Python's scipy library to validate whether the Add-on Total discrepancy between loyal buyers ('Yes') and standard buyers ('No') is statistically significant.

4. Results & Findings

4.1. Shipping Fulfillment Optimization (RQ1)

The Non-dominated Sorting Genetic Algorithm evaluated competing objectives: minimizing total daily shipping costs while maximizing customer satisfaction (weighted by loyalty status) across all completed orders in the dataset.

A critical success factor for this operations model is real-world feasibility. Warehouses operate under strict daily limits for premium packing and shipping. The model's outputs were validated against four capacity constraints for premium shipping methods. The algorithm strictly adhered to all parameters without violating warehouse capacities, proving that the mathematical formulation is viable for daily supply chain operations.

In a Multi-Objective Optimization problem, there is no single "perfect" answer, but rather a Pareto frontier of optimal trade-offs. To provide a singular, actionable recommendation for warehouse managers, the model extracted the "Balanced Compromise Solution" using the `argmin()` function for the crowding distance formula, the solution mathematically closest to the ideal theoretical center of the Pareto front.

Shipping Method	Assigned Orders	Capacity	Slack (Capacity - Assigned)	Constraint Satisfied
Standard	138	Unlimited	Unlimited	True
Expedited	58	100	42	True
Express	4	60	56	True
Overnight	0	30	30	True
Same Day	0	20	20	True

Table 2. Balanced Compromise Solution

The metrics for this optimal compromise for the 200 orders are:

- Projected Daily Shipping Cost: \$1,660.00
- Projected Customer Satisfaction Score: 1,032.5

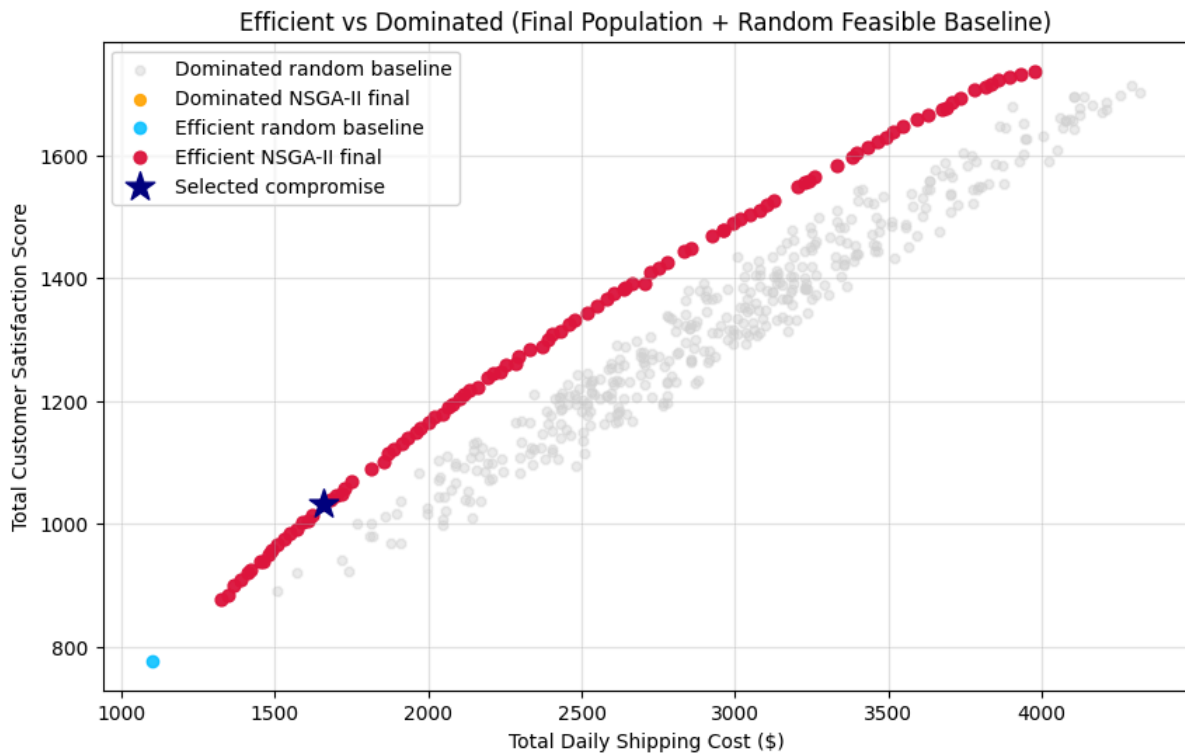


Fig. 3. Pareto front plot

The model exhibits highly rational economic behavior. It completely avoided **Same Day** (\$50) and **Overnight** (\$30) shipping, despite having ample warehouse capacity for them. Instead, it relied heavily on **Expedited** shipping (\$15), utilizing over half of that tier's capacity, while keeping the majority of orders on **Standard** (\$5).

This proves that given the defined business parameters, ultra-premium shipping tiers suffer from severe **diminishing returns**. The \$35 to \$45 premium to upgrade an order to Overnight or Same Day does not yield a proportionally large enough spike in the satisfaction score to justify the expense. **Expedited shipping (\$15)** acts as the operational "sweet spot," generating the most efficient satisfaction ROI per dollar spent.

4.2. Price Elasticity of Demand across different SKUs (RQ2)

This analysis estimates the price elasticity of demand across five major electronic product categories: Tablets, Laptops, Smartphones, Smartwatches, and Headphones. To isolate the effect of price on purchase quantity, the log-log model controlled for seasonality (month of purchase), customer demographics (Age, Loyalty Status), and product quality (Customer Rating).

The table below summarizes the elasticity coefficients and their respective P-values. In a log-log model, the coefficient represents the percentage change in quantity demanded given a 1% change in price.

Product Type	Elasticity	P-Value	Observations	Statistical Significance
Tablet	-0.0184	0.458	2,745	Not Significant
Laptop	+0.2757	0.0004	2,686	Highly Significant ($p < 0.01$)

Smartphone	+0.0097	0.310	4,004	Not Significant
Smartwatch	-0.0439	0.355	2,636	Not Significant
Headphones	+0.1196	0.159	1,361	Not Significant

Table 3. Summary of analysis results

For Tablets, Smartphones, Smartwatches, and Headphones, the P-values are > 0.05 . This means there is no statistically significant relationship between the price of the SKU and the quantity customers purchase in a single order. Demand is practically rigid across different price tiers. Laptops are the only category with a statistically significant result (P-value = 0.000394), but the elasticity is positive (+0.275). This implies that a 10% increase in price correlates with a 2.7% *increase* in the quantity demanded.

The overall lack of negative elasticity across the board suggests that within this retail environment, customers are not simply hunting for the cheapest product. When they log on to buy a smartphone or a tablet, their decision on *how many* to buy is largely divorced from the price tier of the item. Product features, brand reputation, and specific consumer needs are overpowering baseline price sensitivity.

4.3. Customer Segmentation and Loyal Status Evaluation (RQ3)

The unsupervised K-Means machine learning algorithm successfully partitioned the customer base into four distinct strategic cohorts based on their Recency, Frequency, and Monetary (RFM) behaviors. The algorithm utilized standardized inputs to prevent the high-variance monetary axis from dominating the distance calculations.

Cluster ID	Recency	Frequency	Monetary
0	80.82	1.11	2968.57

1	245.11	1.15	3218.16
2	87.55	2.19	7105.41
3	64.17	3.34	14394.39

Table 4. Summary of Cluster Centroids

Cluster 3 ("Champions") represents the most valuable customers. They have the most recent purchase history (lowest recency), the highest transaction frequency, and an exceptionally high aggregate spend. These customers are highly engaged and price-insensitive. Cluster 2 ("Loyal/Growth Customers") exhibits solid multidimensional behavior, purchasing roughly twice within the dataset timeframe and generating healthy revenue. They have shopped relatively recently (within the last 3 months). This is the prime target for up-selling and cross-selling campaigns. They have proven brand affinity but have not yet maximized their lifetime value (CLV) compared to Cluster 3. Customers in Cluster 0 ("Recent One-Time Buyers") have shopped recently but are characterized by low frequency (mostly single-transaction buyers) and low monetary value. They represent a critical conversion opportunity. Post-purchase nurturing campaigns, targeted discounts on lower-friction ancillary items, and onboarding them into the broader retail ecosystem are essential to prevent them from churning. Finally, Cluster 1 ("Hibernating") is the highest-risk group. With a massive average recency of over 8 months (245 days) and stagnant frequency/spend vectors, these customers have effectively abandoned the company. They require aggressive win-back campaigns, deep discounts, or surveys to understand why they churned.

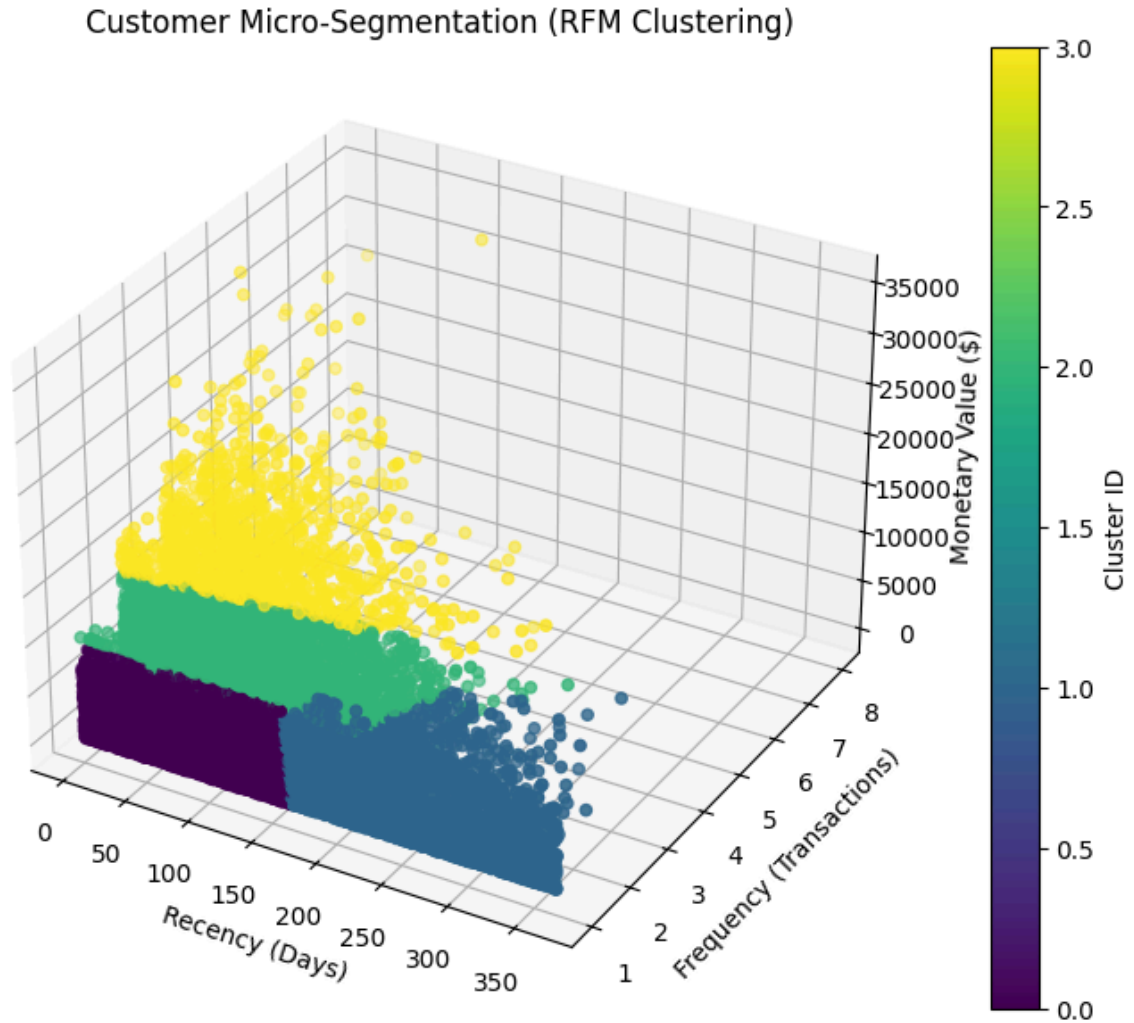


Fig. 4. 3D Scatter of RFM Clusters

For the evaluation of the Loyalty Member variable, a hypothesis is formed, that loyalty ecosystems generate "points pressure," inherently gamifying the purchasing process and driving up high-margin add-on purchases (Chandrasekhar et al., 2024). The dataset was subjected to a rigorous Welch's Independent Two-Sample T-Test to validate this. Below are the statistical findings of the test:

- Mean Add-on Total (Loyalty Members): \$62.31
- Mean Add-on Total (Standard Buyers): \$62.23
- T-Statistic: 0.0820
- P-Value: 0.93466

The p-value of 0.9347 is vastly greater than the standard alpha threshold of 0.05. Therefore, we must reject the initial hypothesis and conclude that there is no statistically significant difference in add-on spending between loyalty program members and non-members.

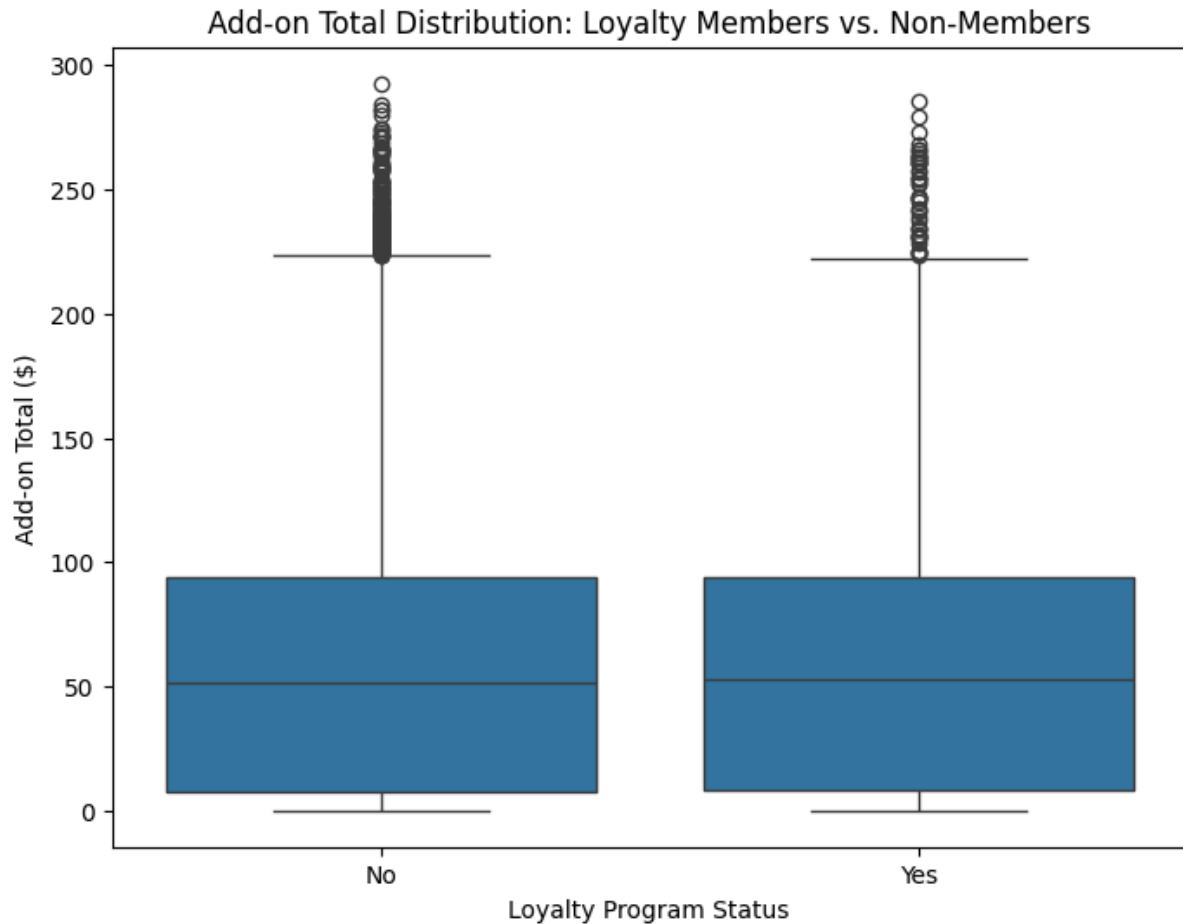


Fig. 5. Loyalty Member vs. Add-on Total

This finding indicates that the current Loyalty Program is highly ineffective at driving secondary, high-margin consumer behavior. While loyalty members might be utilizing the program for core device discounts, the program is completely failing as a strategic lever to subsidize thin device margins with extensive add-on chains.

5. Recommendations

The synthesis of marketing analytics, supply chain operations, machine learning, and multi-objective mathematical optimization provides an interconnected solution for navigating

the immense complexities of modern electronics retail. From all above-mentioned results, the following recommendations are proposed for the company and other relevant stakeholders to implement for the best results of operational and supply chain efficiency and customer retention to maximize profitability.

5.1. Shipping Fulfillment Optimization (RQ1)

Based on the optimization findings, the following actions are recommended for the Operations and Supply Chain team. First, all the constants, such as shipping costs, warehouse capacity and base satisfaction scores, were assumed to simulate a real-world scenario and are not from the dataset. In this scenario, the model actively avoids assigning Same Day and Overnight deliveries due to poor cost-to-satisfaction efficiency. Thus, the cost of premium-tier shipping methods such as Same Day fulfillment should be lowered. The business should utilize this model with its own set of constants to effectively modify the cost of such premium shipping methods.

Warehouse managers should not be locked into a single, rigid fulfillment rule. The generated Pareto front should be deployed as a "daily executive menu." On days with tight cash flow, managers can select a point further down the curve (lower cost, lower satisfaction). During peak promotional events where brand image and speed are critical to preventing order cancellations, they can select a point higher up the curve.

5.2. Price Elasticity of Demand across different SKUs (RQ2)

The slight positive elasticity in laptops and the inelasticity in other categories indicate strong pricing power. The retailer does not need to engage in aggressive "race-to-the-bottom" (continuously lower prices, standards,... in a desperate attempt to undercut its competitors) pricing to drive volume, particularly for high-end electronics.

Since unit price variations don't significantly boost the quantity of the primary item ordered, the retailer should focus on increasing Average Order Value (AOV) through cross-selling and bundling (e.g., offering discounts on accessories only when a premium laptop or smartphone is purchased).

5.3. Customer Segmentation and Loyal Status Evaluation (RQ3)

The statistical analysis proved that loyalty members do *not* spend more on add-ons than non-members. The program is failing its primary objective of driving high-margin add-on revenue. Thus, the company should consider offering loyalty members experiential benefits that cost the company less but hold high perceived value, such as expedited shipping, priority customer service, or extended return windows.

For "Champions", customers exhibiting the highest frequency and monetary spend, these buyers are price-insensitive and highly engaged; offering them discounts needlessly cannibalizes revenue. Instead, they should be nurtured through VIP treatment and early access to flagship products. Conversely, the "Recent One-Time Buyers" cohort represents the highest-leverage conversion opportunity. Post-purchase drip campaigns and targeted discounts on high-frequency, lower-friction items should be aggressively deployed to establish a habit of returning to the store. Finally, "Hibernating" customers, who have been dormant for extended periods, require targeted win-back initiatives or qualitative surveys rather than standard promotional blasts, which are virtually guaranteed to be ignored.

Since mass marketing is highly inefficient when customer behaviors are this heavily stratified, the company should allocate the majority of the promotional budget toward Customer Acquisition Cost (CAC) for new buyers, and conversion campaigns aimed specifically at turning the "Recent One-Time Buyers" into repeat purchasers. The margin saved by eliminating discounts for the "Champions" cohort should be reinvested directly to these targeted acquisition and win-back campaigns.

6. Conclusion

This comprehensive analysis demonstrates the immense strategic value of integrating machine learning, mathematical optimization, and robust marketing analytics to untangle the complexities of the modern electronics retail sector. By systematically interrogating 20,000 historical transaction records, this report successfully isolated distinct inefficiencies within the company's current logistics network, pricing architecture, and customer retention programs.

The findings fundamentally challenge standard operational assumptions. The data proves that funneling capital into ultra-premium shipping tiers yields diminishing returns, and that deploying rigid, discount-heavy pricing strategies is unnecessary for consumer electronics that exhibit strong price inelasticity. Furthermore, the revelation that the current loyalty ecosystem completely fails to stimulate secondary, high-margin purchasing behavior underscores an urgent need for organizational realignment.

Ultimately, mass marketing and static operational rules have been superseded by the need for dynamic, micro-targeted strategies. By adopting the data-driven recommendations outlined in this report, implementing dynamic shipping fulfillment menus, shifting focus toward cross-selling for higher Average Order Value, and executing highly stratified customer lifecycle campaigns, the company can efficiently optimize resource allocation, maximize Customer Lifetime Value (CLV), and secure a resilient competitive advantage.

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Appendix

Use of Gen AI Declaration:

Gemini was used to aid in grammar and sentence structure

Gemini was used in order to help brainstorm ideas in the initial planning stages

Gemini was used to help strengthen certain arguments and aid in developing ideas

Gemini were used to help find reputable sources to strengthen the report